Prayas JEE AIR (2025)

Advanced

Weekly Test 26

DURATION: 180 Minutes | **DATE : 04/01/2025** | **M. MARKS: 180**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **180**.

SECTION-1 (Maximum Marks: 24)

This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value of **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If **ONLY** the correct integer is entered;

  Zero Marks : 0 In all other cases.

SECTION-2 (Maximum marks: 24)

- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

  Negative Marks : –2 In all other cases.

SECTION-3 (Maximum marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Five entries** (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

  Negative Marks : –1 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

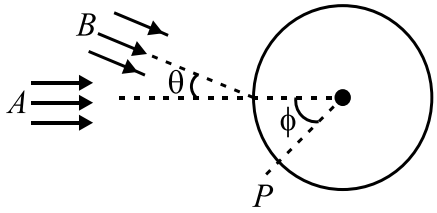
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PART-I (PHYSICS)

SECTION-1

Numerical Value Type Questions

1. Two identical beams *A* and *B* of plane coherent waves of the same intensity and wavelength λ fall on cylindrical screen. The angle between the directions of the beam propagations is θ. Consider a point *P* on the screen at angular position ϕ from the beam *A* as shown in the figure. Find the closest integer to distance in nanometers between adjacent interference fringes on the screen near the point *P*. Assume the distance β between adjacent fringes is much less than the radius of the cylinder. (Given: $\theta = 60°$, $\phi = 15°$, $\lambda = \sqrt{2}$nm)

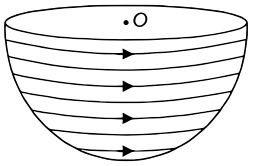


2. A cylindrical long wire of radius *R* carries current I_0 uniformly distributed in its cross-sectional surface. The pressure developed due to magnetic force in wire at distance $\frac{R}{2}$ from its axis is given by $\frac{n\mu_0 I_0^2}{16\pi^2 R^2}$. Find value of *n*.

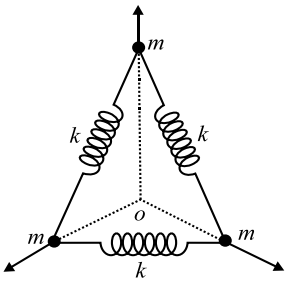
3. Interference fringes were produced in Young's double-slit experiment using light of wavelength 5000 Å. When a film of thickness 2.5×10^{-3} cm was placed in front of one of the slits, the fringe pattern shifted by a distance equal to 20 fringe-width. Find the refractive index of the material of the film.

4. All electron approaches a fixed proton from a large distance with a kinetic energy of 12.1 eV. The electron gets captured by the proton to form a hydrogen atom in an excited state and a photon of wavelength λ_1 = 800Å is emitted. Later, the hydrogen atom de-excites by emitting a single photon of wavelength λ_2. The value of λ_2 (in nm) is. [Take hc = 1240 eV-nm]
Write the answer round off to the two decimal places.

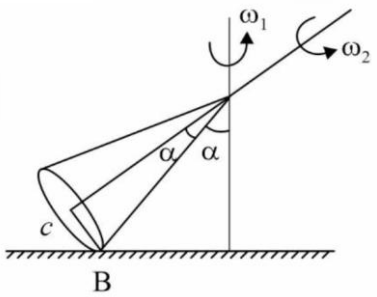
5. A wire is wrapped tightly without any gap over a hemispherical bowl of radius *R*. If *N* is the total number of turns and *i* is the current per turn, the magnetic field at the centre *O* will be $\alpha\frac{\mu_0 Ni}{R}$.
Find the value of α.



6. Three identical ideal springs, each of force constant *k* are joined to three identical balls (each of mass *m*) as shown in the figure. *O* is centroid of the triangle. Initially, each of the springs is in tis natural length. Now all the three balls are simultaneously given small displacements of equal magnitude along the radially outwards directions as shown in the figure and released. The oscillation frequency of each ball is $\frac{1}{\pi}\sqrt{\frac{\beta k}{m}}$. Find the value of β.



7. A uniform solid cone (height $h = 4\,\text{m}, \alpha = 30^\circ$) is inclined against a vertical axis as shown in the figure. The cone rotates about its own axis as well as rotates about the vertical axis with angular speeds marked in the diagram. If the cone does not slip at point B; find $\frac{\omega_1}{\omega_2}$.

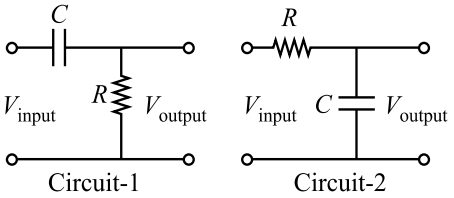


8. Least count of a vernier calliper is 0.01 cm. When the two jaws of the instrument touch each other the 5th division of the vernier scale coincide with a main scale division and the zero of the vernier scale lies to the left of the zero of the main scale. Further more while measuring the diameter of a sphere, the zero mark of the vernier scale lies between 2.4 cm and 2.5 cm and the 6th vernier division coincides with a main scale division. Calculate the diameter (in cm) of the sphere.

SECTION-2

One or More Than One Correct Type Questions

9. Figure shows two circuits (i.e., circuit 1 and circuit 2). In each case V_0 denotes peak value of input voltage. Input voltage is given by $V_{input} = V_0 \sin \omega t$ and output voltage by $V_{output} = V_M \sin(\omega t + \delta)$



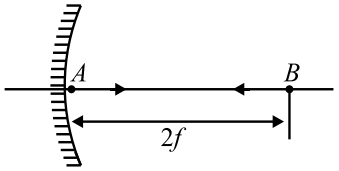
(A) In circuit-1 maximum output voltage is $V_M = \dfrac{V_0}{\sqrt{1+(\omega RC)^{-2}}}$

(B) In circuit-2 maximum output voltage is $V_M = \dfrac{V_0}{\sqrt{1+(\omega RC)^{2}}}$

(C) In circuit-1 average power dissipated is given by $P_{av} = \dfrac{V_0^2}{2R\left[1+(\omega RC)^{-2}\right]}$

(D) Phase angle δ for circuit-2 is given by $|\tan\delta| = \omega RC$

10. At time $t = 0$, two point objects A and B respectively are at pole and centre of curvature of a fixed concave mirror of focal length f; the velocity vectors of A and B are always $\vec{V}_A = u\hat{i}$ and $\vec{V}_B = -u\hat{i}$ respectively, where $\hat{i}$ is unit vector along principal axis directed from pole towards focus and u is a positive constant



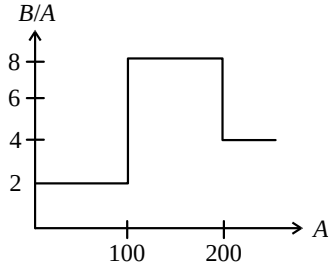
(A) The distance between images of A and B will be $4f$ at time $t = \dfrac{f}{2u}$

(B) Magnitude of relative velocity of image of A and image of B at $t = 0$ is $2u$

(C) Starting from $t = 0$ and before the particles come in contact, distance between image A and B increases

(D) Starting from $t = 0$ and before the particles come in contact, distance between A and B first increases and then decreases

11. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below.



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy

(B) Fusion of two nuclei with mass number lying in the range of $51 < A < 100$ will release energy

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments

(D) Fission of nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

12. A particle of mass 1 kg moves in x-y plane such that its potential energy at a general point (x, y) is defined as $\phi = 3x + 4y$ (in joules). In this conservative force field a particle is released at (6, 4) at $t = 0$, from rest. Choose the correct statements from options given below. (Assume SI units)

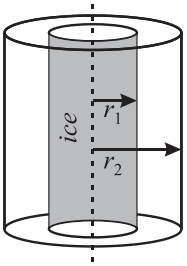
(A) During its motion inside the field the particle moves with a constant non-zero acceleration

(B) The work done by the conservative forces from position of rest of the particle to the instant of the particle crossing x axis is 25 J

(C) During its motion when the particle's trajectory crosses y-axis it has its speed equal to 10 m/s

(D) After 4 seconds from release the particle had its position at coordinates (–18, –28)

13. A 100 cm long cylindrical flask with inner and outer radius $r_1 = 2$ cm and $r_2 = 4$ cm respectively, is completely filled with ice at 0°C as shown in the figure. The constant temperature outside (surrounding) the flask is 40°C. Assume heat exchange occurs only through the curved surface of the flask and water (from melting ice) always remains at 0°C until all the ice melts. (Thermal conductivity of the flask is 0.693 W/m°C, L_{ice} = 80 cal/gm and 1kcal = 4200 J). (Assume that density of ice and water is same as 10^3 kg/m^3)



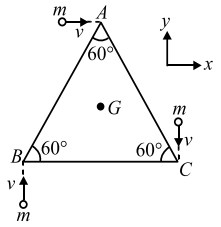
(A) Rate of heat flow from surrounding to the flask is 80π J/s.

(B) Rate of heat flow from surrounding to the flask is 40π J/s.

(C) The rate at which ice melts is $\frac{\pi}{4200}$ Kg / s .

(D) Time taken for ice to completely melt is 28 min.

14. A triangular wedge *ABC* of mass *m* and sides 2*a* lies on a smooth horizontal plane as shown. Three point masses of mass *m* each strike the wedge at *A*, *B* and *C* with speeds *v* as shown. After the collision, the particles come to rest. Select the correct alternative(s).



(A) The centre of mass of *ABC* remains stationary after collision

(B) Then centre of mass of *ABC* moves with a velocity *v* along *x*-axis after collision

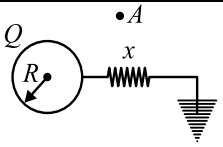
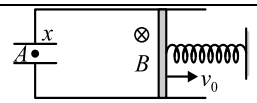
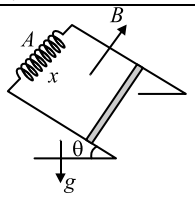
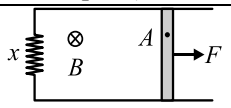
(C) The triangular wedge rotates with an angular velocity $\omega = \frac{2\sqrt{3}mva}{I}$ about its centre of mass (here, *I* is the moment of inertia of triangular wedge about its centroid axis perpendicular to its plane)

(D) A point lying at a distance of $\left(\frac{a}{2\sqrt{3}}\right)$ from *G* on perpendicular bisector of *BC* (below *G*) is at rest just after collision

SECTION-3

Matrix Match Type Questions

15. List-I contains variation of some physical quantities and List-II contains some system at $t = 0$ in which a circuit element *x* and a point *A* is marked. Choose the correct match(s) from List-II for List-I.

	List-I		**List-II**
I	Current through *x* varies sinusoidally with time	P	 A conducting sphere is given charge *Q* and connected to earth through resistance *x* (as shown).
II	Field (either magnetic or electric) at *A* decreases with time.	Q	 One end of two conducting smooth rails are connected to a capacitor and a sliding wire is given a velocity v_0 in a uniform magnetic field *B*. The sliding wire is connected to a spring of spring constant *k* and initially the spring is relaxed (as shown) (*A* is point inside capacitor space)
III	Potential difference across *x* increases with time.	R	 Two conducting rails are fixed on an inclined plane of inclination θ their one ends are connected to an inductor and a uniform magnetic field exist perpendicular to the inclined plane. A conducting wire slides on the rails without friction under gravity (as shown) (*A* is point inside inductor space)
IV	Field (either magnetic or electric) at *A* varies sinusoidally	S	 Two conducting rails are connected by a resistance *x* and a straight conducting wire slides on the rails under the action of constant external force *F* perpendicular to the wire. A uniform magnetic field *B* exist perpendicular to the plane (as shown) (*A* is point inside wire)
		T	None of the above

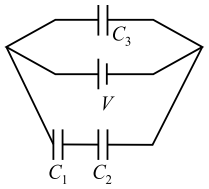
(A) I→R; II→P; III→Q,R; IV→T

(B) I→Q,R; II→P; III→S; IV→Q,R

(C) I→Q; II→R; III→Q,R; IV→R

(D) I→S; II→T; III→R; IV→P,R

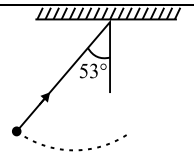
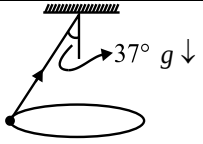
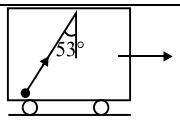
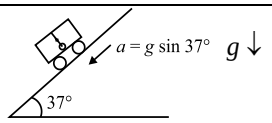
16. In the circuit shown in figure plates of capacitor C_2 are gradually pulled apart. Then match the following:



	List-I		**List-II**
I	Potential difference across C_1	P	Will remain same
II	Energy stored in C_3	Q	Will increase
III	Potential difference across C_2	R	Will decrease
IV	Electric field between the plates of capacitor C_1	S	Less than V
		T	None of these

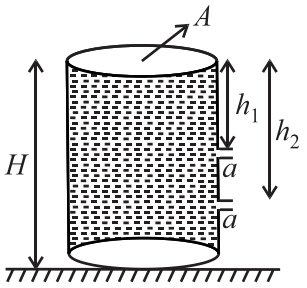
(A) I→R; II→Q; III→T; IV→P
(B) I→Q,S; II→R,S; III→P; IV→R
(C) I→T; II→P,S; III→Q; IV→S
(D) I→R,S; II→P; III→Q,S; IV→R

17. As shown a small bob of mass m, tied at the end of a light string of length l is put into different situations with other end of string tied to fixed ceiling or roof of a moving object. The list-II has entries corresponding to the tension in the string for situations mentioned in list-I. Match the two list. (g is acceleration due to gravity)

	List-I		**List-II**
I	 Extreme position in simple pendulum	P	$\frac{4mg}{5}$
II	 Conical pendulum	Q	$\frac{5mg}{4}$
III	 Cart is accelerated such that particle remains at rest as shown w.r.t. cart	R	$\frac{3mg}{5}$
IV	 Cart is accelerating freely on the incline. Particle remains at rest w.r.t. cart as shown	S	$\frac{5mg}{3}$
		T	$\frac{4mg}{3}$

(A) I→S; II→P; III→R IV→Q
(B) I→R; II→Q; III→S; IV→P
(C) I→P; II→Q; III→T; IV→S
(D) I→R; II→P; III→T; IV→Q

18. A massless container is filled with a liquid of density $1200\ \text{kg/m}^3$. Two holes each of area 'a' are made at a depth of $h_1 = 5$ cm, $h_2 = 15$ cm from the top of the container. Initial height of the liquid in the container is $H = 20$ cm. Now match the following $\left[\frac{A}{a} = 10\right]$



	List-I		**List-II**
I	Initial acceleration (in m/s^2) of the container with upper hole open and lower hole closed. (nearly)	P	1
II	Initial acceleration (in m/s^2) of the container with lower hole open and upper hole closed. (nearly)	Q	$\frac{3}{2}$
III	Initial acceleration (in m/s^2) of the container with both hole open. (nearly)	R	$\frac{1}{2}$
IV	Initial ratio of horizontal range from the two holes. (nearly)	S	2
		T	$\frac{5}{2}$

(A) I→Q; II→R; III→S; IV→P
(B) I→R; II→Q; III→S; IV→P
(C) I→R; II→P; III→T; IV→S
(D) I→S; II→T; III→R; IV→Q

PART-II (CHEMISTRY)

SECTION-1

Numerical Value Type Questions

19. From the following reaction

(a) $CH_3-\overset{\overset{O}{\|}}{C}-H + xCl_2 \xrightarrow{NaOH} Cl_3CH + H-\overset{\overset{O}{\|}}{C}-ONa$

(Haloform reaction)

('x' is minimum possible integer)

(b) $y\ H-\overset{\overset{O}{\|}}{C}-H + zNH_3 \longrightarrow$ one mole Urotropine (Use to treat chronic urinary tract infections) (y & z are minimum possible intgers)

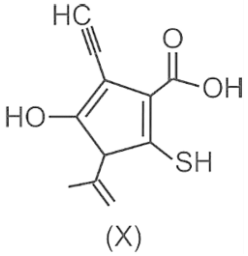
Find out $\left[\dfrac{x+y}{z}\right]$

20. Total number of substance which contain hexagonal planar rings in their structures is:
Graphite, $(BN)_x$, $B_3N_3H_6$, C_6H_6, B_2H_6, $H_3P_3O_9$

21. A gaseous substance remains partially dissociated $A_n(g) \rightleftharpoons nA(g)$. If the given sample of gas has $A_n(g)$ 52% dissociated at equilibrium and the rate of diffusion of pure oxygen is 1.25 times that of equilibrium mixture of A under identical condition, determine 'n'. [Atomic weight of A is 32 and O is 16 amu]

22. A 100 mL solution is prepared by mixing 8.2 g of Na_3PO_4 and 4.5625 g of HCl. If Ka_1, Ka_2 and Ka_3 of H_3PO_4 are 10^{-5}, 10^{-8} and 10^{-13} respectively then the pH of the prepared solution is

23. How many moles of marsh gas are liberated when excess of methyl magnesium bromide is treated with one mole of the given compound X?



(X)

24. An optically active substance, A, decomposes into optically active substances 'B' and 'C' as:

$$A \xrightarrow{K=0.001\,min^{-1}} 2B + C$$

The specific rotations of A, B and C are +40°, +10° and –30° per mole, respectively. If initially A and C were present in 4 : 3 mole ratio, the time (in min), after which the sample becomes optically inactive, is_____. (ln 2 = 0.7, ln 5 = 1.6, ln 7 = 2.0, ln 13 = 2.5)

25. Gold metal is extracted from its ore by treating the crushed rock with an aerated cyanide solution. The unbalanced equation for the reaction is

$Au(s) + CN^-(aq) + O_2(g) \rightarrow Au(CN)_2^-(aq)$. Use any of the following data at 27°C to calculate $\Delta G°$ for this reaction in basic solution. (Considering cell reaction with 4 moles of electrons.)

Given K_f for $Au(CN)_2^-$ = 10^{38}

K_w of H_2O = 10^{-14}

$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(\ell)\ \ E° = 1.2V$

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)\ \ E° = 1.5V$

$Au^{3+}(aq) + 2e \rightarrow Au^+\ \ E° = 1.5V$

Use equation $\Delta G° = -2RT\log_{10} K$ and R = 8J K^{-1} mol^{-1}, F = 96500 C. Give your answer (magnitude only) in kJ.

26. Among the following, total number of compound(s) which produce oxygen as the only gaseous product on gentle heating, is/are_____.
$NaNO_2$, KNO_3, Ag_2O, $Ca(NO_3)_2$, $KClO_3$, HgO, CaO, Na_2O, $NaNO_3$, $Ba(NO_3)_2$, H_2O_2.

SECTION-2

Multiple Correct Type Questions

27. Which of the following statements is/are correct about the hybridisation and the atomic orbitals involved in that hybridization?

(A) In dsp^3 hybridisation (trigonal bipyramidal geometry) s, p_x, p_y, p_z and d_{z^2} orbitals of outer most shell are involved

(B) In sp^3d^2 hybridisation (octahedral geometry) s, p_x, p_y, p_z and outer d_{z^2}, $d_{x^2-y^2}$ orbitals are involved

(C) In dsp^2 hybridisation (Tetrahedral geometry) s, p_x, p_y and $d_{x^2-y^2}$ orbitals are involved

(D) In sp^3d^3 hybridisation (pentagonal bipyramidal geometry) s, p_x, p_y, p_z and outer d_{xy}, $d_{x^2-y^2}$ and d_{z^2} orbitals are involved

28. The incorrect statement(s) about the following reaction sequence is/are

$$\text{Cumene } (C_9H_{12}) \xrightarrow[\text{ii. } H_3O^+]{\text{i. } O_2/hv} \underset{\text{aromatic compound}}{[P]} \xrightarrow[\text{iii. } H^+]{\substack{\text{i. NaOH}\\ \text{ii. } CO_2}} \underset{\text{Major}}{[Q]} + \underset{\text{Minor}}{[R]}$$

$$[Q] \xrightarrow{CH_3OH / H_2SO_4} \underset{\text{Major}}{[S]}$$

(A) [Q] and [S] are example of positional isomers

(B) [R] on treatment with Acetic anhydride in presence of acid gives aspirin

(C) pKa value of [Q] is more than [R]

(D) [S] gives dark violet coloration with 1% aqueous $FeCl_3$ solution

29. A gas X is passed through water to form a saturated solution. The aqueous solution on treatment with $AgNO_3$ gives white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas Y. Select the correct statement(s)

(A) Y is H_2

(B) X is Cl_2

(C) X is CO_2

(D) Y is CO

30. 1.0 g of Mg is heated strongly in a vessel containing 0.5 g of O_2 to form MgO. Which of the following statements is/are correct?

[Atomic mass: O = 16, Mg = 24]

(A) 1.5 g MgO is formed in the reaction

(B) 1.25 g MgO is formed in the reaction

(C) O_2 is completely consumed in the reaction

(D) 0.25 g Mg is left unreacted at the end of reaction

31. Which of the following is/are correct?

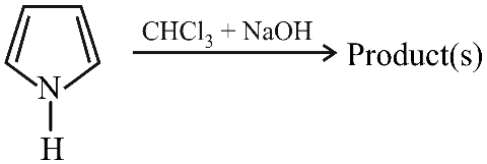
(A) $XeF_6 + SiO_2 \longrightarrow XeOF_4 + SiF_4$

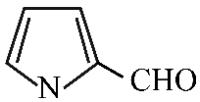
(B) $XeOF_4 + H_2 \xrightarrow{300^\circ C} Xe + H_2O + HF$

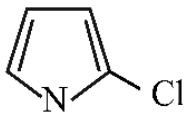
(C) $XeO_2F_2 + H_2O \rightarrow XeO_3 + HF$

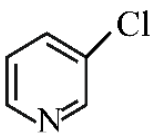
(D) XeO_3 in strongly basic medium disproportionate to form perxenate ion and elemental Xe.

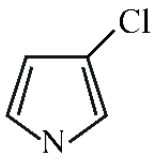
32. In the following reaction, the product (s) formed is/are



(A) 

(B) 

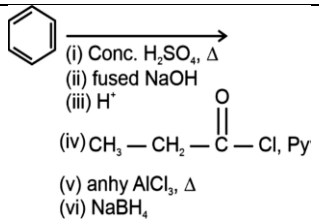
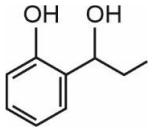
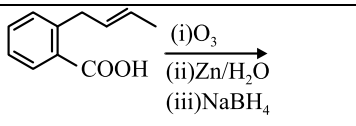
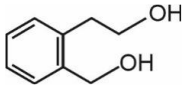
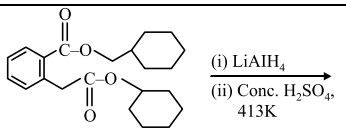
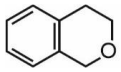
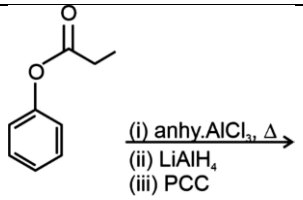
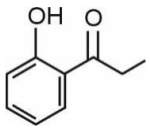
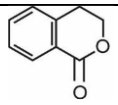
(C) 

(D) 

SECTION-3

Matrix Match Type Questions

33. Match the starting materials and reagents given in List-I with the final products formed in List II

	List-I		**List-II**
I		P	
II		Q	
III		R	
IV		S	
		T	

(A) I→P; II→T; III→R; IV→S

(B) I→P; II→Q; III→R; IV→T

(C) I→R; II→T; III→R; IV →P

(D) I→Q; II→P; III→Q; IV→S

34. List-I lists various processes and List-II contains different conditions true for these processes. Match the List-I with List-II.

	List-I		**List-II**
I	Isothermal reversible compression of an ideal gas	P	$\Delta U = 0$
II	Adiabatic reversible compression of an ideal gas	Q	$\Delta H = 0$
III	Adiabatic expansion against a constant external pressure of an ideal gas	R	$W = 0$
IV	Vaporisation of liquid at normal boiling point	S	$\Delta S_{sys} > 0$
		T	$\Delta S_{total} = 0$

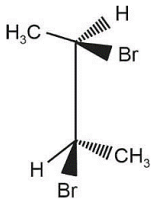
(A) I→ P,Q,T; II→T; III→S; IV→S, T

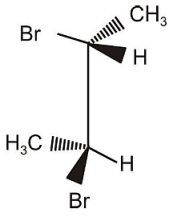
(B) I→P,Q; II→R; III→P, R; IV→S,T

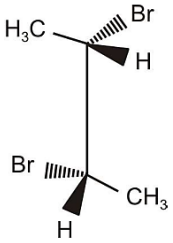
(C) I→Q,R; II→S; III→P,R; IV→T

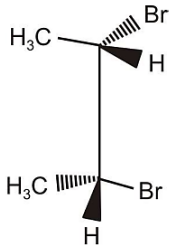
(D) I→T; II→Q,R; III→P,Q; IV→P,Q,S

35. Consider the following structures.

(a) 

(b) 

(c) 

(d) 

List-I contains a pair of molecules out of a, b, c and d and List-II contains relationship between them. Match List-I with List-II

	List-I		**List-II**
I	a and b	P	Identical
II	c and d	Q	None is meso
III	a and c	R	Diastereomers
IV	b and d	S	Enantiomer
		T	Both are optically active

(A) I→R; II→R; III→Q,S,T; IV→P

(B) I→P; II→Q; III→Q,S,T; IV→R

(C) I→Q; II→R; III→Q,R,T; IV→P

(D) I→R; II→Q; III→P,R,T; IV→P

36. List-I contains complex(es) and List-II contain different characteristics and types of hybridization.

	List-I		**List-II**
I	$[Co(en)_3]^{3+}$	P	sp^3d^2 hybridisaion
II	$[Co(ox)_3]^{3-}$	Q	Diamagnetic
III	$[Co(H_2O)_6]^{2+}$	R	d^2sp^3 hybridisation
IV	$[Co(NO_2)_6]^{3-}$	S	Paramagnetic
		T	Chelate ligand

Identify the option having correct match?

(A) I→P,R,T; II→Q,R,T; III→P,R; IV→Q,R

(B) I→Q,R,T; II→Q,R,T; III→P,S; IV→Q,R

(C) I→P,R,T; II→Q,R,T; III→P,T; IV→P,R

(D) I→R,S,T; II→Q,R,T; III→P,Q; IV→Q,R

PART-III (MATHEMATICS)

SECTION-1

Numerical Value Type Questions

37. The minimum value of $9\sec^2 x + 4\operatorname{cosec}^2 x$ is

38. If $\vec{x}$ and $\vec{y}$ are two non zero, non-collinear vectors satisfying $\left((a-3)\alpha^2+(b-4)\alpha+(c-1)\right)\vec{x}+\left[(a-3)\beta^2+(b-4)\beta+(c-1)\right]\vec{y}+\left[(a-3)\gamma^2+(b-4)\gamma+(c-1)\right](\vec{x}\times\vec{y})=\vec{O}$ (where α, β, γ are three distinct numbers), then the value of $a^2 + b^2 + c^2$ is equal to

39. A fair die is rolled four times. The probability that each of the final three rolls is at least as large as the roll preceding it may be expressed in the form $\frac{m}{n}$ where m and n are relatively prime positive integers. The value of $m + n$ is

40. Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2 = 3A + \alpha I$. *If* $A^4 = 21A + \beta I$, then $\alpha - \beta$ is

41. Suppose $f : R \to (0, \infty)$ be a differentiable function such that $5f(x + y) = f(x) \cdot f(y), \forall x, y \in R$. If $f(3) = 320$, the $\sum_{n=0}^{5} f(n)$ is equal to k, then $\frac{k}{25}$ is

42. The area of the region bounded by $x = -1$, $x = 2$ $y = x^2 + 1$ and $y = 2x - 2$ is

43. If $I = \int_1^4 \left(\{x\}\right)^{[x]} dx$, then $\frac{24}{13} I$ is equal to _____.

where $[\cdot]$ denote greatest integer function and $\{\cdot\}$ denote fractional part 'x'

44. The number of solution(s) of equation $\cos^{-1}x = \sin^{-1}\left(\frac{-1}{4}\right)$ is

SECTION-2

One or More Than One Correct Type Questions

45. If $\left(9+\sqrt{80}\right)^n = I + f$ where I, n are integers and $0 < f < 1$, then:

(A) I is an odd integer

(B) I is an even integer

(C) $(1 + f)(1 - f) = 1$

(D) $1 - f = \left(9-\sqrt{80}\right)^n$

46. Let $f(x) = \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 1$ and $g(x) = \int_1^x (2t^2 - \ln t) f(t)\, dt$ $(x > 1)$, then:

(A) g is increasing on $(1, \infty)$

(B) g is decreasing on $(1, \infty)$

(C) g is increasing on (1, 2) and decreasing on $(2, \infty)$

(D) g is decreasing on (1, 2) and increasing $(2, \infty)$

47. Let ΔPQR be formed by the common tangents to the circle $x^2 + y^2 + 6x = 0$ and $x^2 + y^2 - 2x = 0$, then the which of the following is (are) true?

(A) centroid of ΔPQR is (1, 0)

(B) radius of circle inscribed in ΔPQR is 1

(C) Area of ΔPQR is $3\sqrt{3}$

(D) the y-intercept of common tangent having negative slope is 3.

48. If $\sin^2 x + \sin x = (a + 2)$, then which of the following statement(s) is(are) correct?

(A) Number of integral values of a for real solution to exist is 3.

(B) There exists no solution for $a < \frac{-9}{4}$ or $a > 0$.

(C) The minimum value of a for real solution is -2.

(D) Number of positive prime values of a for real solution to exists is 1.

49. Number of ways in which 30 identical things are distributed among six persons is:

(A) ${}^{17}C_5$ if each gets odd number of things

(B) ${}^{16}C_{11}$ if each gets odd number of things

(C) ${}^{14}C_5$ if each gets even number of things (excluding 0)

(D) ${}^{15}C_{10}$ if each gets even number of things (excluding 0)

50. Consider $f(x) = x\left[x^2\right] + \frac{1}{\sqrt{1-x^2}}$

(A) $f(x)$ is an even function

(B) $f(x)$ is an odd function

(C) $f(x)$ is periodic

(D) Range of $f(x)$ has only positive values

SECTION-3

Matrix -Match Type Questions

51. Match List-I to List-II according to the given condition and choose the correct option.

	List-I		**List-II**
I	Tangents at the ends of focal chord of a parabola	P	Meet at directrix
II	Let P is a point on the parabola and tangent at P meets the directrix at Q. If S is the focus of parabola then SP and SQ	Q	Meet at an angle of 90°
III	If $A\left(at_1^2, 2at_1\right)$ and $B\left(at_2^2, 2at_2\right)$ are the ends of a focal chord of $y^2 = 4ax$, then	R	$t_1t_2 = -1$
IV	If the normal at $\left(at_1^2, 2at_1\right)$ meets the parabola $y^2 = 4ax$ again at $\left(at_2^2, 2at_2\right)$, then	S	$t_2 = -t_1 - \frac{2}{t_1}$
		T	$t_1t_2 + t_1^2 + 2 = 0$

(A) I→P,Q; II→Q; III→R; IV→S,T

(B) I→P,S; II→Q; III→R; IV→S,T

(C) I→P,Q; II→R; III→S; IV→S,T

(D) I→P,R; II→S; III→Q; IV→S,T

52. Match the following:

	List-I		**List-II**
I	The differential equation of the family of curves $y = e^x(A\cos x + B\sin x)$, where A, B are arbitrary constants, has the degree n and order m. Then, the values of n and m are.	P	1, 3
II	The degree and order of the differential equation of the family of all parabola whose axis is the x-axis, are.	Q	1, 1
III	The order and degree of the differential equations of the family of circles touching the x-axis at the origin are.	R	2, 2
IV	The degree and order of the differential equation of the family of ellipse having the same foci are.	S	1, 2
		T	3, 2

(A) I→S; II→S; III→S; IV→S;

(B) I→S; II→S; III→Q; IV→S;

(C) I→Q; II→S; III→R; IV→P;

(D) I→S; II→Q; III→T; IV→S;

53. A variable plane cuts the x-axis, y-axis and z-axis at the points A, B and C respectively such that the volume of the tetrahedron $OABC$ remains constant equal to 32 cubic unit and O is the origin of the coordinate system

	List-I		**List-II**
I	The locus of the centroid of the tetrahedron is	P	$xyz = 24$
II	The locus of the point equidistant from O, A, B and C is	Q	(0, 24, 0)
III	If A (2, 0, 0) and B (0, 2, 0) then C is	R	$xyz = 30$
IV	If A (4, 0, 0) and C (0, 0, 2) then B is	S	(0, 0, 48)
		T	$xyz = 3$

(A) I→P; II→R; III→Q; IV→S
(B) I→T; II→P; III→S; IV→Q
(C) I→Q; II→R; III→Q; IV→S
(D) I→T; II→P; III→Q; IV→R

54. Match the equation in z, in List-I with the corresponding values of arg(z) in List-II

	List-I (Equation in z)		**List-II (Principal value of arg (z))**
I	$z^2 - z + 1 = 0$	P	$-2\pi/3$
II	$z^2 + z + 1 = 0$	Q	$-\pi/3$
III	$z^2 + 1 = 0$	R	$\pi/2$
IV	$z^2 - 1 = 0$	S	$2\pi/3$
		T	0

(A) I→Q; II→P,S; III→R; IV→T
(B) I→S,R; II→S,P; III→Q,S; IV→S,R
(C) I→R,P; II→Q,S; III→R; IV→T
(D) I→R,P; II→Q,P; III→Q,S; IV→P,R



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